KISHORE RAM G

Robotics Engineer

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PROFESSIONAL SUMMARY

Robotics and Automation undergraduate with hands-on experience in ROS 2 (Jazzy), Gazebo, RViz2, OpenCV, and autonomous mobile robot development. Experienced in URDF modeling, ROS 2 communication, computer vision, and sensor integration. Familiar with Python, C++, and Linux, with a strong interest in robotics software development and intelligent autonomous systems.

INTERNSHIP

Industrial Automation & PLC Internship, Saravana Electronics | 15 Days Rajapalyam

- Learned the fundamentals of PLCs, ladder logic, and industrial automation systems.
- Gained practical exposure to industrial control panels, wiring, field instrumentation, and panel design.
- Understood industrial automation workflows by working with Software, Hardware, R&D, and Panel Design teams.

Drone Technology Internship, NSIC Technical Services Centre | 7 Days Chennai

- Learned the fundamentals, applications, and architecture of UAVs and agricultural drones.
- Assembled the mechanical structure of an agricultural drone and understood its major components.
- Gained practical knowledge of drone calibration procedures and UAV system integration.

EDUCATION

Bachelor of Engineering: Robotics & Automation 09/2023 – Present

Dhaanish Chennai College of Engineering | Chennai

TECHNICAL SKILLS

Robotics

- ROS 2(Jazzy), Gazebo, RViz2, URDF, TF2, OpenCV

Programming

Python ,C++

IoT & Embedded Systems

Microcontrollers, Sensor Interfacing, System Integration

Design & Tools

Fusion 360

PROJECTS

Autonomous Mobile Robot (SLAM & Navigation)

- Developed an autonomous mobile robot using ROS 2 Jazzy, Gazebo, and RViz2.
- Implemented SLAM Toolbox for real-time mapping using a LiDAR sensor.
- Integrated the Nav2 stack for autonomous path planning and goal-based navigation.
- Built the robot model using URDF/Xacro and tested it in a simulated environment.

Technologies: ROS 2 Jazzy | SLAM Toolbox | Nav2 | Gazebo | RViz2 | URDF/Xacro | LiDAR | Python

Gesture Control Robot

- Developed a hand gesture-controlled robot using MediaPipe and OpenCV.
- Implemented real-time hand landmark detection to control robot movements.

Technologies: Ros2 Jazzy, Gazebo, Rviz, Python, OpenCV, MediaPipe

Autonomous Obstacle Avoidance Robot

- Developed an autonomous obstacle avoidance robot using ROS 2 and Gazebo.
- Integrated LiDAR and camera sensors for obstacle detection and navigation.
- Implemented obstacle avoidance logic and validated the robot in a simulated environment.

Technologies: ROS 2 Jazzy, Gazebo, LiDAR, Camera, Python

Leader–Follower Robot using ROS 2

- Developed a multi-robot leader–follower system using ROS 2 Jazzy, Gazebo, and RViz2.
- Implemented LiDAR-based autonomous obstacle avoidance for the leader robot.
- Developed a vision-based follower robot using OpenCV and ArUco marker detection for real-time target tracking.
- Designed both robots using Xacro/URDF and validated autonomous leader–follower behavior in a complex Gazebo environment.

Technologies: ROS 2 Jazzy, Gazebo, RViz2, OpenCV, ArUco, LiDAR, Camera, URDF, Xacro, Python

Autonomous Line Follower Robot (ROS 2)

- Developed an autonomous line-following robot using ROS 2 Jazzy, Gazebo, RViz2, and OpenCV.
- Implemented camera-based line detection and centroid-based steering control.
- Designed the robot using URDF and validated it in Gazebo simulation.

Technologies: ROS 2 Jazzy, Gazebo, RViz2, OpenCV, Python